HOMEWORK ASSIGNEMENT

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1
TASK 1 : SINGLE COMPARISON

1.1 Question 1

We first visualize the data to better understand the distribution of fasting glucose levels in both groups.

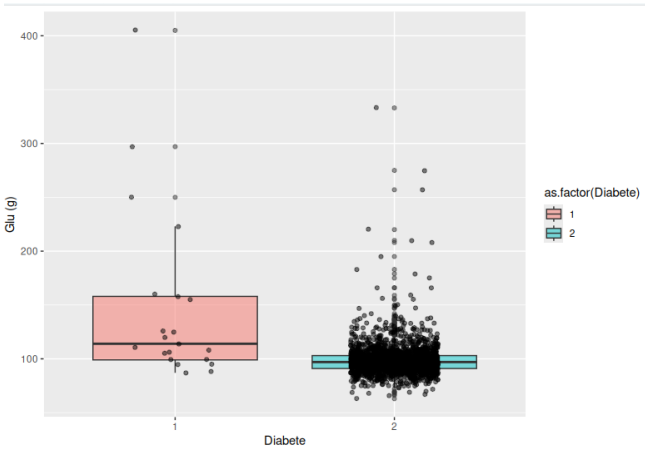

Figure 1 – Boxplot of fasting glucose levels (Diabetic vs Healthy respondents)

The boxplot suggests a substantial difference in central tendency between the two groups. We also observe that diabetic respondents represent a much smaller proportion of the sample compared to healthy respondents.

Since the two groups are independent and sample sizes are unequal, we perform a Welch two-sample t-test (which does not assume equal variances).

The hypotheses are :

$$H_0 : \mu_D = \mu_H$$
$$H_1 : \mu_D \neq \mu_H$$

The observed sample means are :

$$\bar{x}_D = 149, \quad \bar{x}_H = 99.$$

The test yields a p-value of 1.02×10^{-2}.

Since $p < 0.05$, we reject H_0. There is statistically significant evidence that the mean fasting glucose level differs between diabetic and non-diabetic respondents.

1.2 Question 2

The boxplot suggests a difference in glucose levels between adults and seniors.

As the two age groups are independent and variances may differ, we again use a Welch two-sample t-test.

$$H_0 : \mu_{Adult} = \mu_{Senior}$$
$$H_1 : \mu_{Adult} \neq \mu_{Senior}$$

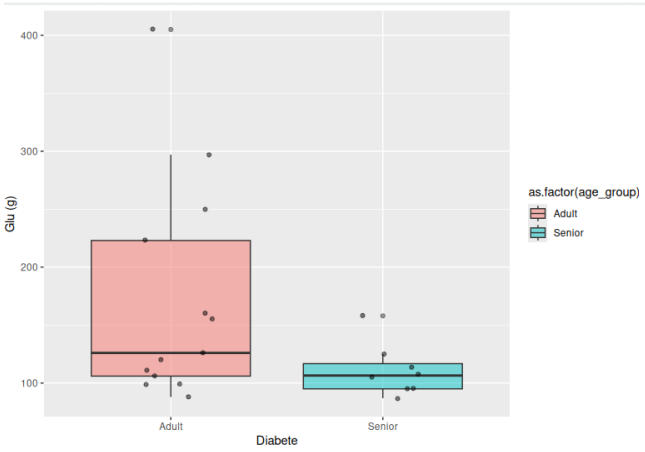

Figure 2 – Boxplot of fasting glucose levels (Adults vs Seniors among diabetics)

The observed sample means are :

$$\bar{x}_A = 172, \quad \bar{x}_S = 111.$$

The test produces a p-value of 4.36×10^{-2}.

Since $p < 0.05$, we reject H_0 at the 5% significance level. There is evidence of a difference in mean fasting glucose levels between adult and senior diabetic respondents.

1.3 Question 3

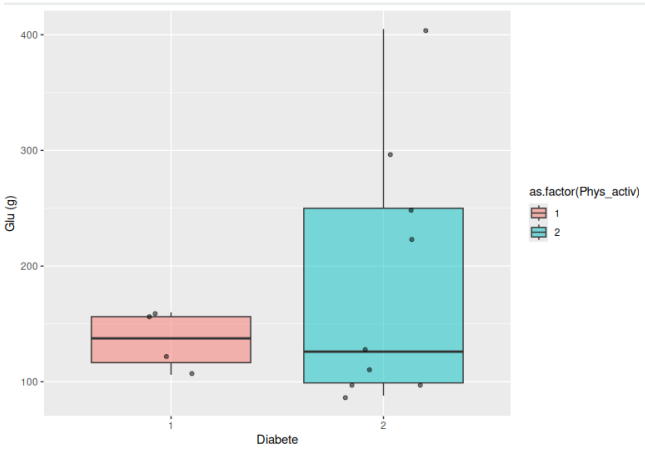

Figure 3 – Boxplot of fasting glucose levels (Vigorous activity vs No vigorous activity)

The boxplot shows only a small apparent difference between the two groups. However, the number of individuals performing vigorous physical activity is extremely small (only 4 observations), which strongly limits the statistical power of the test.

A Welch two-sample t-test gives a p-value of 2.06×10^{-1}.

Since $p > 0.05$, we fail to reject H_0. There is no statistically significant evidence of a difference in mean glucose levels. However, this result should be interpreted with caution due to the very small sample size in the physically active group.

1.4 Question 4

The observed counts are :
— Women : 8 diabetic, 1072 non-diabetic
— Men : 13 diabetic, 1127 non-diabetic
We test :

$$H_0 : p_{Male} = p_{Female}$$

$$H_1 : p_{Male} \neq p_{Female}$$

Due to the small number of diabetic cases, we first perform Fisher's exact test, which is appropriate for small counts. This test yields a p-value of 3.84×10^{-1}.

We also perform a two-sample proportion test, which gives a p-value of 4.51×10^{-1}.

Both tests lead to the same conclusion : we fail to reject H_0 at the 5% significance level. There is no statistically significant evidence that the proportion of diabetic respondents differs between males and females.

Fisher's exact test is more reliable in this context because it does not rely on large-sample approximations, unlike the proportion test.

Conclusion

The aim of the exercise was to find a correlation between diabetic traits and other physical traits such as gender or level of blood glucose after fasting. We found that being diabetic and being older had an impact on the level of blood glucose after fasting but we failed to reject an independence between the gender and being diabetic and between level of glucose and having a physical job, mainly because the number of data was lacking.

2
TASK 2 : GENE EXPRESSION DATA

The liver dataset contains measurements of rat liver toxicity (cholesterol levels) and the expression levels of 3116 genes (log-ratios between two experimental conditions). The goal is to identify genes linked to the response variable, cholesterol.

2.1 Most Correlated Gene and Linear Model

We first computed the Pearson correlation between cholesterol and each of the 3116 gene expression levels. The gene with the maximum absolute correlation was identified as `A_42_P631473`, with a correlation of approximately 0.732, indicating a strong positive linear association.

We fitted a linear model of the form $Y = \beta_0 + \beta_1 X + \varepsilon$, where Y is cholesterol and X is the expression level of the selected gene. The estimated slope was about 64.86 and the intercept 86.46. The F-test for the model yielded a p-value of 6.31×10^{-12}, far below the conventional 0.05 threshold. We therefore reject the null hypothesis of no relationship : there is strong evidence of a significant linear association between this gene's expression and cholesterol level. The model explains approximately 53.6% of the variance in cholesterol ($R^2 \approx 0.536$).

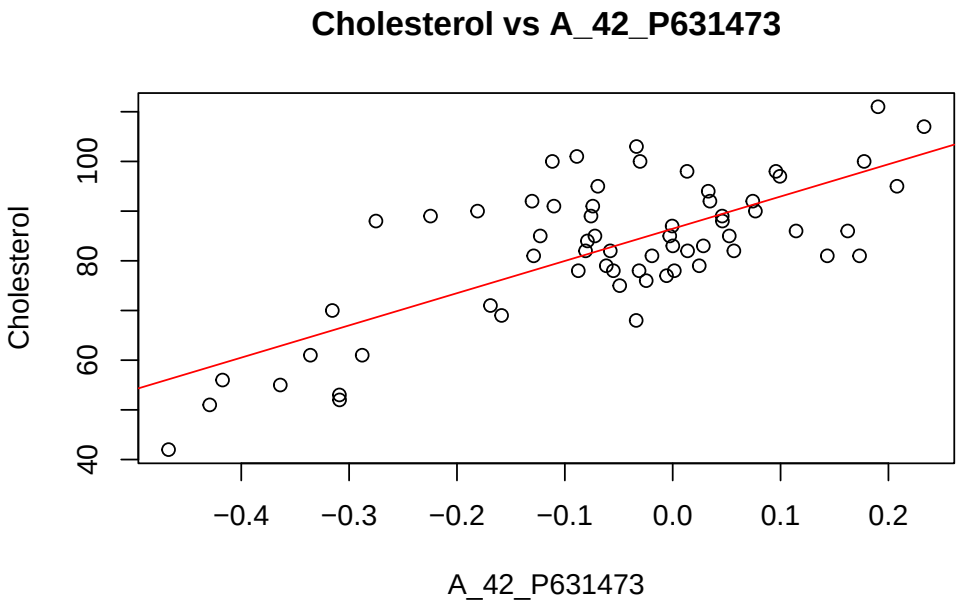

Figure 4 – Cholesterol as a function of the expression level of gene A_42_P631473, the gene most correlated with cholesterol. The red line is the fitted linear regression.

2.2 Genome-Wide Testing

We extended the analysis to all 3116 genes by fitting a separate linear model for each gene, with cholesterol as the response and gene expression as the covariate. For each model, we extracted the p-value of the t-test for the slope (testing $H_0 : \beta_1 = 0$). This produced 3116 p-values, one per gene.

2.3 P-Value Distribution and Rank Plot

The p-values were ordered and plotted against their ranks (1 to 3116). On the same plot, we added the line $y = x/m$, where $m = 3116$ is the number of tests. Under the global null hypothesis (no gene is associated with cholesterol), the ordered p-values would be expected to follow the uniform distribution, and the empirical cumulative distribution would approximately coincide with $y = x/m$.

In our data, the ordered p-values lie well below this line for small ranks, indicating that many genes have very small p-values. This suggests that a non-negligible proportion of genes are truly associated with cholesterol, rather than all associations being due to chance. The departure from the diagonal line provides a qualitative assessment of the presence of signal in the data.

2.4 False Discovery Rate Control (Benjamini-Hochberg)

To identify genes linked to the response while controlling the expected proportion of false discoveries (False Discovery Rate, FDR) at 5%, we applied the Benjamini-Hochberg procedure. Let $p_{(1)} \leq p_{(2)} \leq \ldots \leq p_{(m)}$ be the ordered p-values. The procedure finds the largest k such that $p_{(k)} \leq (k/m) \cdot \alpha$ with $\alpha = 0.05$, and rejects the hypotheses corresponding to $p_{(1)}, \ldots, p_{(k)}$.

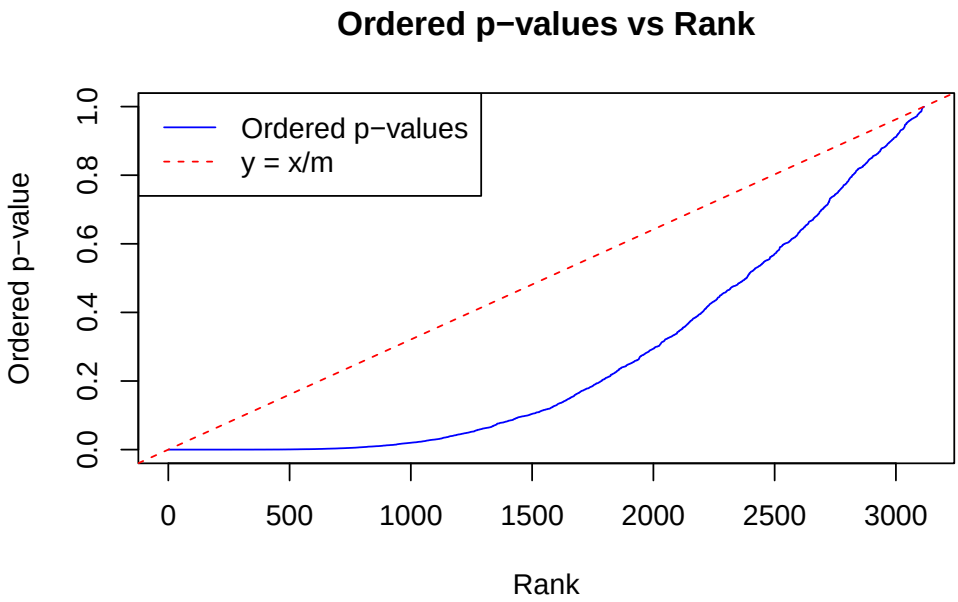

Figure 5 – Ordered p-values as a function of rank. The blue curve shows the empirical ordered p-values ; the red dashed line is $y = x/m$ (with $m = 3116$), corresponding to the uniform null.

Applying this procedure, we obtained **941 discoveries** : 941 genes are declared significant with FDR $< 5\%$. Among these, we expect at most about 5% to be false positives.

2.5 Family-Wise Error Rate Control (Bonferroni)

For a more conservative guarantee—controlling the probability of making at least one false discovery (Family-Wise Error Rate, FWER) at 5%—we used the Bonferroni correction. Under this approach, we reject the null hypothesis for gene i if $p_i \leq \alpha/m$, where $m = 3116$ and $\alpha = 0.05$. The Bonferroni threshold is therefore $0.05/3116 \approx 1.60 \times 10^{-5}$.

With this criterion, we obtained **220 discoveries**. This is fewer than the 941 genes from the FDR procedure, as Bonferroni is designed to strongly limit the chance of any false discovery, at the cost of reduced power and more false negatives.

2.6 Conclusion

The gene expression analysis successfully identified genes associated with cholesterol in the liver dataset. Gene `A_42_P631473` showed the strongest correlation. Depending on the desired trade-off between sensitivity and specificity : the Benjamini-Hochberg procedure yields 941 genes (FDR $< 5\%$) for broader exploration ; the Bonferroni procedure yields 220 genes (FWER $< 5\%$) for a more conservative, high-confidence set.

3
TASK 3 : NON PARAMETRIC REGRESSION

3.1 Question 1

The scatter plot in Figure 6 shows a clear nonlinear and decreasing relationship between x and y. The response drops quickly for small values of x and seems to stabilize near zero for larger values. This suggests that a simple linear model would not be appropriate here.

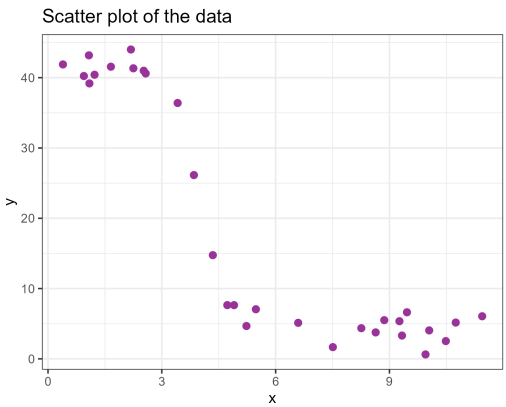

Figure 6 – Scatter plot of y versus x.

3.2 Question 2

We fit polynomial models of degree 0 to 3 to the data. To avoid collinearity issues between the powers of x, we use orthogonal polynomials (`poly(x,d)` in R), which allows us to interpret the t-tests on each coefficient independently.

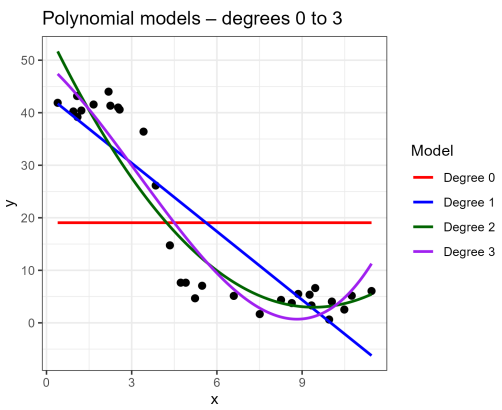

Figure 7 – Polynomial fits of degree 0–3 superimposed on the data.

From Figure 7, we see that the constant and linear models clearly do not capture the shape of the data. The degree-2 and degree-3 curves look very similar visually, so we use statistical criteria to decide between them.

Table 1 – Goodness-of-fit metrics for polynomial models of degree 0–3.

Degree	R^2	Adj. R^2	AIC	BIC
0	0.000	0.000	259.86	262.67
1	0.792	0.784	214.81	219.01
2	0.891	0.883	197.46	203.07
3	0.906	0.895	194.97	201.98

We select the **degree-2 model**. There is a significant improvement when going from degree 1 to degree 2 : the LRT and ANOVA both give $p < 0.05$, and the adjusted R^2 increases from 0.784 to 0.883. Also, we can see that adding a cubic term does not help : the LRT is not significant and BIC increases, which means that the extra complexity is not justified.

The diagnostic plots in Figure 8 look satisfactory : the residuals are roughly centred around zero with no obvious pattern, and the Q-Q plot does not show strong departures from normality.

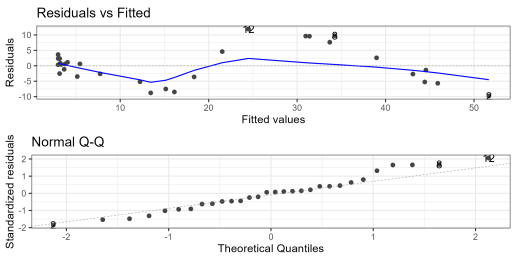

Figure 8 – Diagnostic plots for the degree-2 polynomial model.

One limitation of this model is that a quadratic polynomial eventually increases again for large x, which contradicts the decreasing trend that is visible in the data. This is why it seems interesting to consider a nonlinear model (cf question 3).

3.3 Question 3

We consider two nested exponential models :

$$\mathcal{M}_1 : f(x) = A\,e^{-bx}, \qquad \mathcal{M}_2 : f(x) = A\,e^{-bx} + C$$

Both are fitted by nonlinear least squares using `nls` in R.

Choosing between $\mathcal{M}_1$ and $\mathcal{M}_2$. As we have $\mathcal{M}_1$ nested in $\mathcal{M}_2$ (it corresponds to $C = 0$), we use an ANOVA F-test. The test gives $p = 0.23 > 0.05$, so we fail to reject $H_0 : C = 0$. This is also confirmed by the summary of $\mathcal{M}_2$, where C has a p-value of 0.41. The asymptote parameter is therefore not really justified, so we select $\mathcal{M}_1$.

Comparing $\mathcal{M}_1$ with the degree-2 polynomial. The two models are not nested, so we can rely on AIC and BIC only.

Table 2 shows that the degree-2 polynomial achieves a lower AIC and BIC than $\mathcal{M}_1$. We therefore retain the **degree-2 polynomial** as our final model !

Table 2 – AIC and BIC comparison between the selected nonlinear model and the degree-2 polynomial.

Model	AIC	BIC
Polynomial degree 2	197.46	203.07
$\mathcal{M}_1$ (exponential)	202.84	207.05

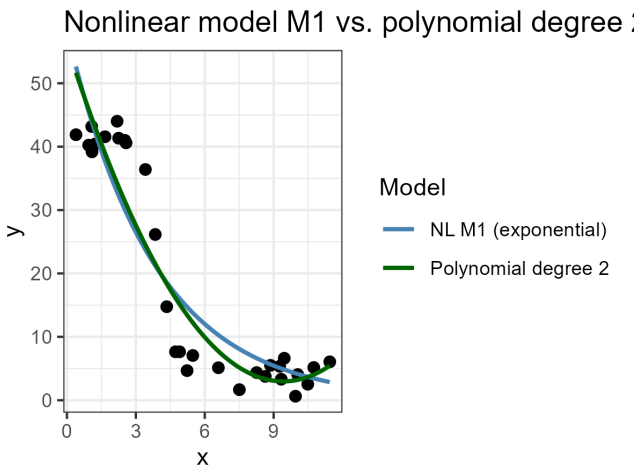

Figure 9 – Nonlinear model $\mathcal{M}_1$ and polynomial degree 2.

3.4 Question 4

Since the degree-2 polynomial was selected as the best model in question 3, we can compute the confidence and prediction intervals for $x_{\text{new}} = (1, \ldots, 10)$ directly using `predict()` in R.

The **confidence interval** for $\mathbb{E}(Y_{\text{new}})$ quantifies the uncertainty on the mean predicted response : it gives the range in which the true mean value of y is likely to lie for a given x. The **prediction interval** for Y_{new} is wider, because it adds the residual variability (it gives the range in which a new individual observation is expected to fall).

Table 3 – 95% confidence (CI) and prediction (PI) intervals for the degree-2 polynomial at $x_{\text{new}} = 1, \ldots, 10$.

x	$\hat{f}(x)$	CI lower	CI upper	PI lower	PI upper
1	45.348	40.828	49.868	32.237	58.459
2	35.873	32.623	39.124	23.144	48.603
3	27.596	24.565	30.627	14.921	40.271
4	20.516	17.136	23.895	7.753	33.279
5	14.633	10.907	18.359	1.774	27.492
6	9.947	6.119	13.776	-2.942	22.836
7	6.460	2.811	10.108	-6.377	19.296
8	4.169	0.865	7.473	-8.574	16.912
9	3.076	-0.084	6.236	-9.631	15.782
10	3.180	-0.634	6.994	-9.705	16.065

We can see that for moderate values of x, the confidence intervals are narrow and the predictions are reliable. However, for $x \geq 6$, the lower bound of the prediction interval becomes negative, and for $x \geq 9$ the confidence interval also crosses zero. This shows once again the known limitation of the degree-2 polynomial : it starts increasing again for large x, which produces unreliable predictions at the boundary of the observed data range.

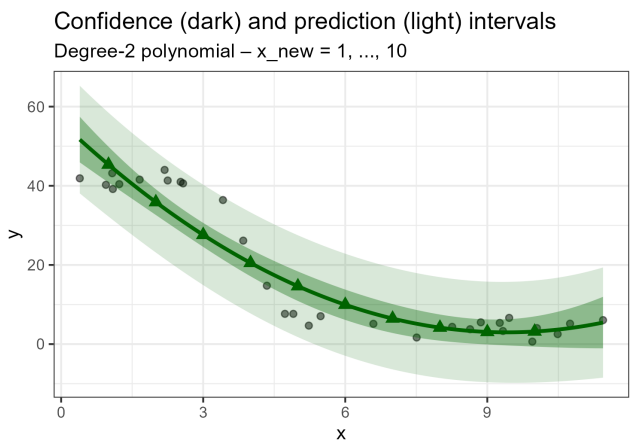

Figure 10 – Fitted degree-2 polynomial with 95% confidence (dark green) and prediction (light green) intervals.

Conclusion

The aim of the exercise was to find the best model to describe the relationship between x and y. The data clearly showed a nonlinear decreasing pattern, which polynomial models of low degree struggle to capture properly.

Among polynomial models, degree 2 was selected based on AIC, BIC and sequential tests (since adding a cubic term brought no significant improvement). We also fitted an exponential nonlinear model, but we showed that the degree-2 polynomial achieved better AIC and BIC values overall.

The main limitation of our final model is its behaviour for large x. Indeed,the quadratic curve eventually increases again, which is inconsistent with the data and leads to prediction intervals crossing zero for $x \geq 6$. A nonlinear model such as $f(x) = Ae^{-bx}$ could be more physically coherent in that regard, even if the information criteria slightly favour the polynomial on this dataset.

4
S&P500 DAILY RETURN

4.1 Data Preparation

The S&P 500 data were loaded from the `sp500\_history.csv` file. The variable of interest, the daily return, was calculated by subtracting the opening value from the closing value (`Close - Open`) for each day studied.

4.2 Modeling with a Normal Distribution

First, we model the daily returns using a standard normal distribution. The empirical estimators for the mean and standard deviation obtained are :

— $\hat{\mu} \approx 0.449$
— $\hat{\sigma} \approx 23.71$

Fit Analysis : Observing the empirical histogram overlaid with the theoretical density shows significant divergences :

- **At the center :** The empirical histogram forms a much higher and narrower peak than the theoretical bell curve, this means that the market stagnates more than the normal distribution predicts.
- **At the extremes :** The theoretical curve flattens out very quickly towards zero. However, the data clearly show the presence of thick tails that could correspond to black day or market crashes .

The single normal distribution model is inadequate. It severely underestimates the probability of extreme events .

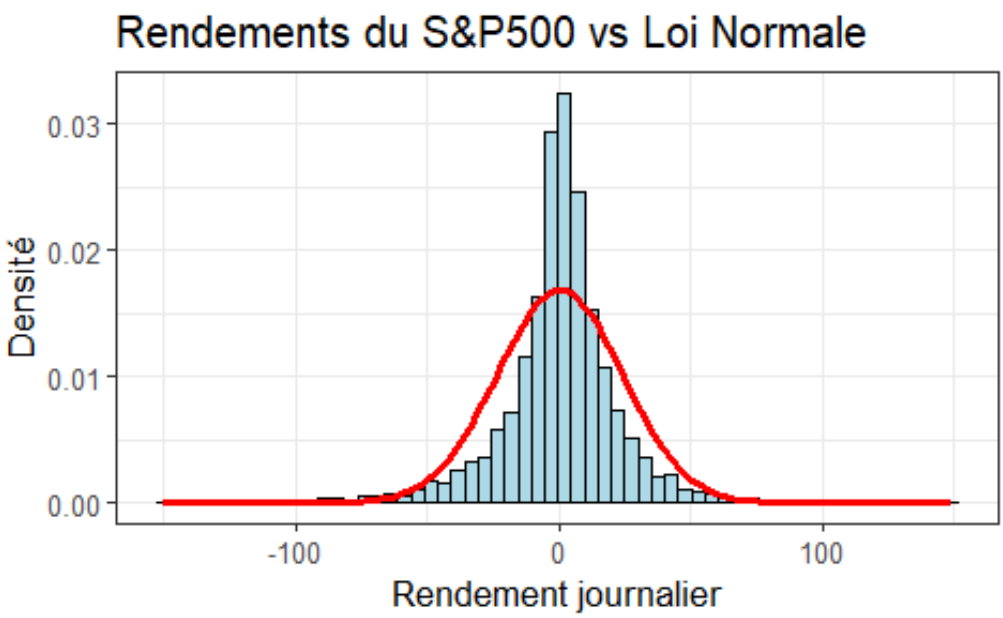

Figure 11 – Fit of a normal distribution to the S&P 500 daily returns. We can clearly observe that the theoretical distribution underestimates the tails.

4.3 Modeling with a Gaussian Mixture

To overcome the limitations of the previous model, we fit normal mixture models for a number of components p ranging from 2 to 6, using the EM algorithm (implemented via the `mixtools` package).

Results Analysis and Convergence :

- The mixture models ($p \geq 2$) largely outperform the simple model ($p = 1$), with their BIC criteria being significantly lower.
- By increasing the number of components, we notice that the EM algorithm diverges or encounters numerical instabilities from $p = 4$ onwards (`WARNING! NOT CONVERGENT!`).
- This divergence is explained by an overfitting phenomenon and a mathematical singularity : the algorithm tends to isolate a few extreme observations in a dedicated component whose variance (σ^2) collapses toward zero.

The minimum of the BIC criterion is reached for the model at $p = 3$ (BIC ≈ 38011.04). This model offers the best compromise by effectively capturing the different volatility regimes of the market.

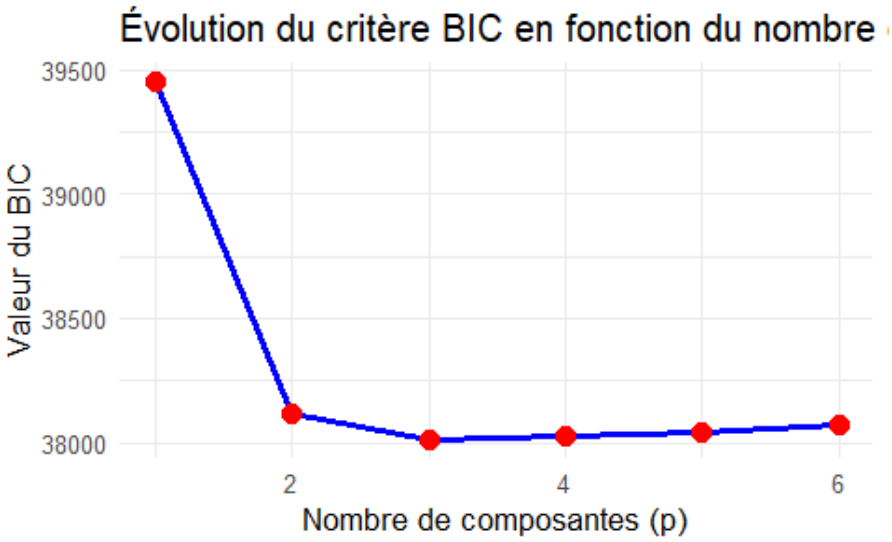

Figure 12 – Evolution of the BIC criterion depending on the number of components p. The minimum is reached for $p = 3$.

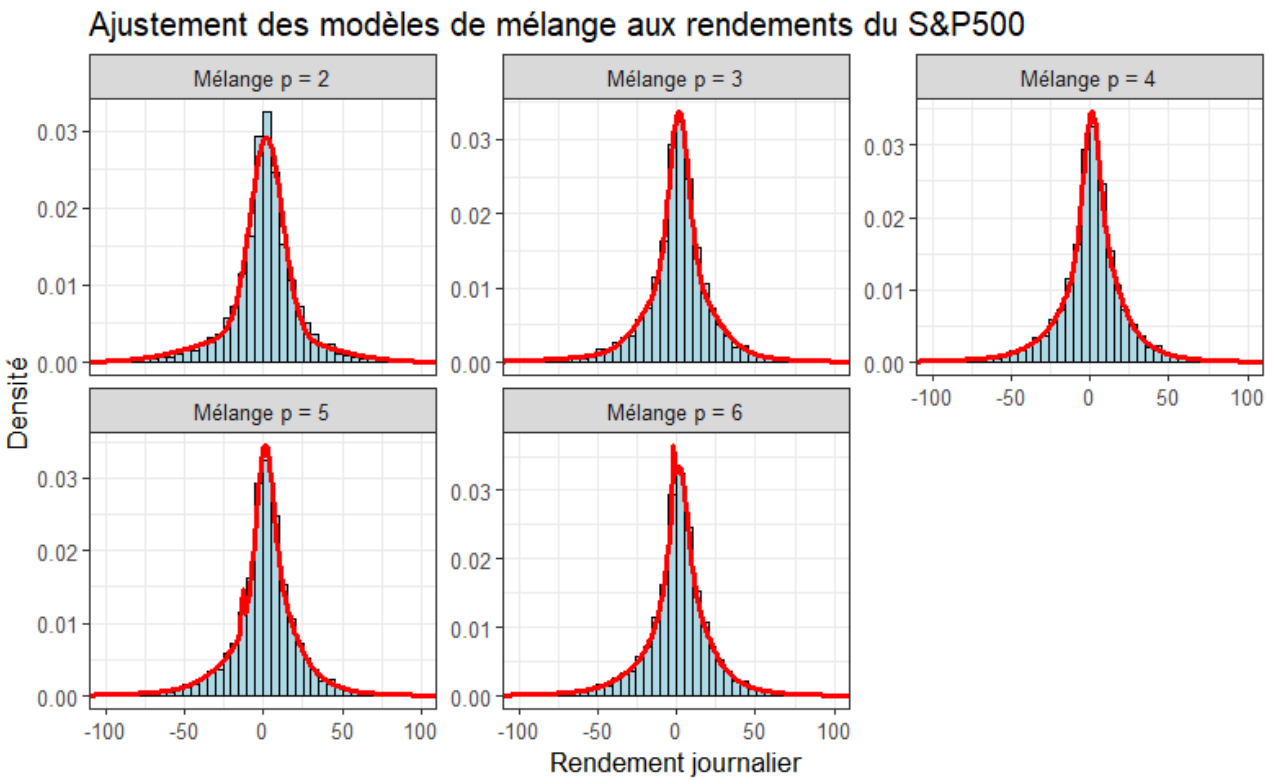

Figure 13 – Densities of the different mixture models. For $p \geq 4$, the algorithm creates degenerate components (extremely sharp peaks), illustrating the overfitting phenomenon.

4.4 Modeling with a Student's t-Distribution (Location-Scale)

Motivation : The main motivation for this model could be to better modelize the tails. The Student's t-distribution has a key parameter : ν (degrees of freedom). The smaller ν is, the wider the tails, allowing for a realistic probability assignment to extreme values using a single mathematical formula.

Algorithm and Estimation : We used a numerical optimization approach via Maximum Likelihood (the `optim` function in R) to minimize the negative log-likelihood of the given density.

The estimated parameters for this model are :

— Degrees of freedom (ν) $\rightsquigarrow$ 2.069

— Location parameter (m) $\rightsquigarrow$ 1.414

— Scale parameter (a) $\rightsquigarrow$ 11.986

The very low value of ν mathematically confirms the presence of extremely fat tails.

4.5 Final Model Selection

To decide between these families of models, our methodology evaluates the tradeoff between the goodness-of-fit (BIC), and algorithmic stability.

1. **Rejection of the normal model ($p = 1$) :** Its BIC (39455.13) is the worst.
2. **Limitations of mixture models :** Although the lowest BIC is reached for $p = 3$ (38011.04), this model requires estimating 8 parameters. Furthermore, it is prone to numerical instabilities (divergence for $p \geq 4$) and the *label switching* problem.
3. **Superiority of the Student's t-distribution :** The Student's model achieves a highly competitive BIC (38090.41), representing a negligible 0.2% difference compared to the best mixture, while using only **3 parameters**. Its estimation is numerically very stable, and it natively captures the fat tails.

Conclusion : Based on the principles of parsimony and stability, the **Student's t-distribution model** is the undisputed winner. It offers the most robust and elegant modeling for the S&P 500 dynamics.

4.6 Test on the Expected Daily Return

To determine if the expected daily return is statistically different from zero, we performed a One-Sample t-test with the null hypothesis $H_0 : \mu = 0$.

The test returns the following results :

— t-statistic $\rightsquigarrow$ 1.2416

— p-value $\rightsquigarrow$ 0.2145

With a p-value of 0.21 (well above the standard significance level $\alpha = 0.05$), **the data do not reject the null hypothesis**. We do not have sufficient statistical evidence to claim that the average return is different from zero. The extreme daily volatility (the noise) completely masks any potential positive average return (the signal).